

Systematic Options: The Wheel Strategy on NSE Derivatives

Research Report — Brindco Quant Researcher Take-Home Assignment

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Scope. This report reads the current Part A stock-wheel outputs from outputs_real/ and Part B synthetic NIFTY-wheel outputs from outputs_nifty/, produced from downloaded NSE equity and derivatives bhavcopies through 2026-06-30. Verified split, bonus, rights, demerger, and symbol events are applied to the stock wheel and equal-weight universe. The action screen and limitations below define the remaining gaps.

1. Overview and verdict

The table below is generated directly from outputs_real/summary.csv. NIFTY 50 uses the official NSE Indices total-return series, which reinvests constituent dividends. The selected-universe benchmark includes recorded dividends and non-cash entitlements.

Series	End value	CAGR	Annual vol	Sharpe	Sortino	MDD	Calmar
Wheel portfolio	INR 36,787,835	9.84%	12.05%	0.32	0.28	-27.74%	0.35
NIFTY 50 total return	INR 42,283,722	12.22%	17.99%	0.38	0.35	-38.27%	0.32
Universe buy-and-hold	INR 66,785,977	20.40%	26.41%	0.61	0.55	-44.15%	0.46

Finding. The stock wheel finished below both benchmarks: 9.84% CAGR versus 12.22% for NIFTY 50 TR and 20.40% for the selected-universe buy-and-hold. It reduced volatility and maximum drawdown relative to both, but its Sharpe ratio remained below NIFTY TR. Premium collection therefore did not translate into superior total or risk-adjusted return. The March 2020 episode confirms the expected short-volatility and left-tail exposure.

NSE Bhavcopy Backtest: Equity Curves

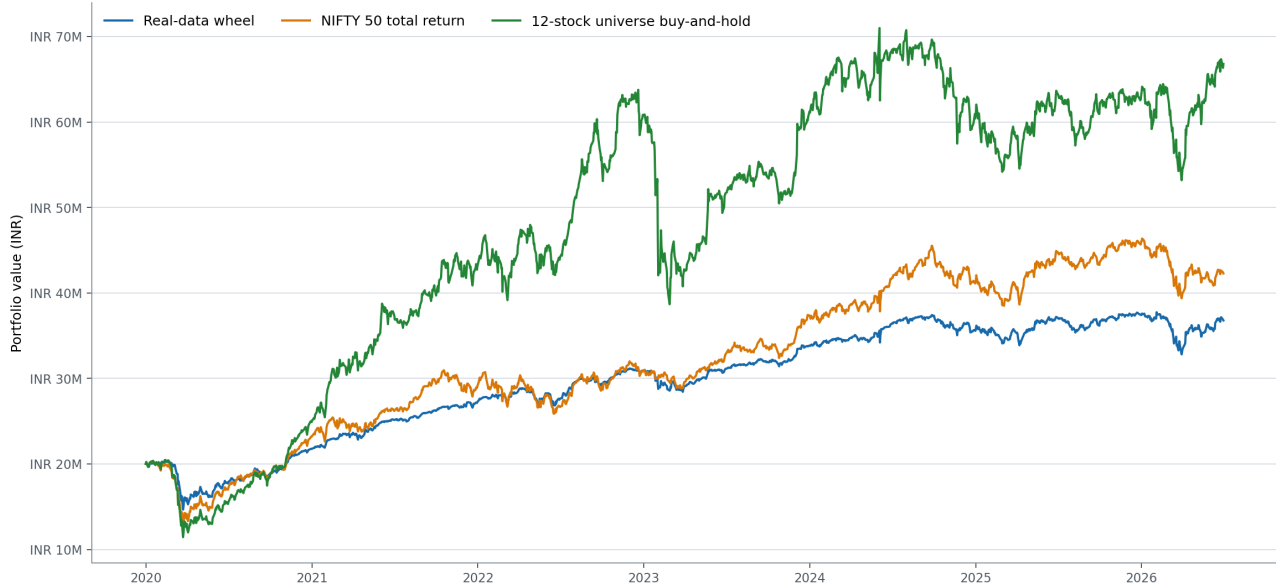


Figure 1. Portfolio equity curves from the real-data run.

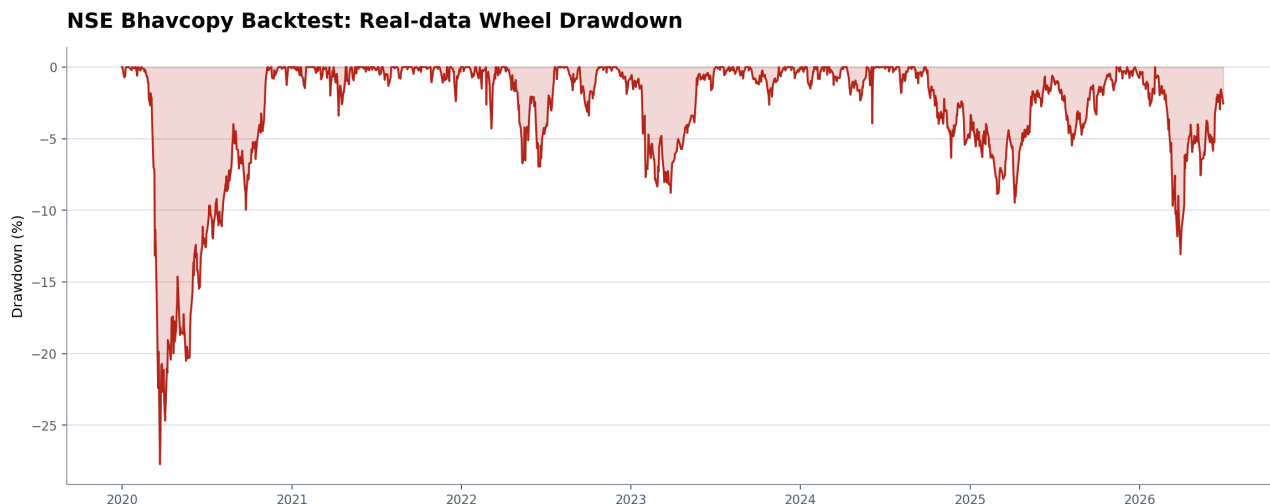


Figure 2. Drawdown calculated from the wheel equity curve.

2. Part A strategy specification and diagnostics

Capital and entry. INR 20 million is split equally across 12 fixed sleeves. Each sleeve waits until the second price date, uses the previous trading day's adjusted underlying close as its strike reference, and models sale at that day's observed option close. A contract needs positive premium, volume and open interest. The nearest listed monthly expiry must be at least seven calendar days away; size is rounded down to whole exchange lots. Cash collateral earns no interest. The 6.5% rate used in Sharpe and Sortino is a performance hurdle, not portfolio income; a yield-bearing collateral alternative is available through --cash-rate but is not the base case because no Treasury-bill or liquid-fund position is modeled.

Wheel rules. With no shares, sell the liquid put nearest 5% below the reference close and reserve the full strike obligation. An ITM put at expiry delivers shares at strike. With shares, sell the call nearest 5% above the reference close, never below the assigned net cost basis; if no qualifying quoted strike exists, sit out. An ITM call delivers the covered shares at strike and any residual shares are sold to reset to cash. Otherwise the expired leg is replaced in the next eligible session. There are no discretionary early profit takes, stops or rolls. Mandatory assignment has delivery costs but no execution slippage; modeled market sales do. A 20% option-margin proxy rises to 35% for ITM deliverables in the final seven calendar days, with pledged shares valued at 80% of close.

The engine recorded **1,681 trade-log events**, including 433 short-put entries, 100 assignments, 404 covered-call entries, and 89 call-aways. It inferred lots for 616 option entries, carried 23 stale marks, and recorded 11 expiry-date revisions.

Part A decision logic

Decision	Rule used by the engine
Reference price	Previous trading-day stock close, transformed to today's ex-action basis
Expiry	Nearest quoted expiry after entry with at least 7 calendar days remaining
Put strike	Listed strike nearest reference x 95%; strike must remain below reference
Call strike	Listed strike nearest reference x 105%, but never below net share basis
Quote filter	Observed option close > 0, volume > 0 and open interest > 0
Execution / mark	Sell at current-day option close; subsequently mark at NSE settlement
Sizing	Put: floor(cash / strike / lot). Call: floor(shares / lot). Whole lots only
Expiry	Stock close below put strike assigns shares; above call strike calls shares away

Example: on 2 January 2020 RELIANCE used the 1 January close of INR 1,509.60. The 5% put target was INR 1,434.12; the nearest eligible strike was INR 1,440, sold at that day's observed INR 7.20 option close. Two 500-share lots fit the sleeve. In the later March assignment, the INR 1,320 put premium of INR 50.55 set net basis to INR 1,269.45; the next call was therefore struck at INR 1,280 even though the simple 5% target was only INR 1,119.51.

Ticker	CSPs	Assign %	Call %	Avg days	Capture	Premium	Costs
RELIANCE	38	21.05%	17.07%	154.6	-5.34%	INR 1,381,618	INR 36,607
HDFCBANK	41	19.51%	19.44%	138.7	-7.32%	INR 912,708	INR 34,529
ICICIBANK	48	12.50%	17.24%	124.8	20.63%	INR 1,376,520	INR 30,118
TCS	32	31.25%	28.12%	84.9	11.39%	INR 1,414,564	INR 55,859

Ticker	CSPs	Assign %	Call %	Avg days	Capture	Premium	Costs
INFY	25	16.00%	6.98%	336.0	-20.53%	INR 847,920	INR 17,828
SBIN	40	25.00%	29.03%	118.3	-49.21%	INR 1,797,964	INR 49,188
AXISBANK	24	37.50%	16.33%	195.6	-67.23%	INR 1,618,914	INR 50,200
KOTAKBANK	33	39.39%	26.67%	105.4	-58.21%	INR 1,357,920	INR 59,792
LT	52	13.46%	28.57%	116.7	7.42%	INR 1,253,359	INR 40,142
TATAMOTORS	26	26.92%	19.35%	157.5	14.88%	INR 1,880,005	INR 42,066
BAJFINANCE	49	26.53%	54.17%	67.9	-56.90%	INR 2,092,000	INR 73,600
ADANIENT	25	20.00%	18.18%	92.8	4.66%	INR 2,359,882	INR 27,806

CSPs counts short-put entries. Avg days covers completed delivery-to-call-away episodes. Premium capture is realized option premium less intrinsic settlement losses, divided by total premium sold; open contracts enter the denominator until expiry.

Per-name risk and return

Ticker	CAGR	Vol	Sharpe	Sortino	MDD	Peak	Trough	Recovery	Calmar
RELIANCE	8.52%	15.27%	0.20	0.18	-24.34%	2020-03-03	2020-03-23	2020-04-22	0.35
HDFCBANK	7.35%	13.47%	0.12	0.11	-21.09%	2020-03-03	2020-03-24	2020-07-08	0.35
ICICIBANK	10.36%	17.21%	0.29	0.25	-31.61%	2020-03-03	2020-03-23	2020-11-13	0.33
TCS	4.95%	15.54%	-0.02	-0.02	-44.92%	2024-12-13	2026-06-30	not yet	0.11
INFY	0.52%	18.31%	-0.23	-0.21	-36.30%	2025-01-10	2026-06-30	not yet	0.01
SBIN	11.57%	13.69%	0.41	0.34	-29.30%	2020-02-26	2020-05-22	2021-02-04	0.39
AXISBANK	14.06%	26.27%	0.40	0.36	-52.30%	2020-03-03	2020-03-24	2021-01-08	0.27
KOTAKBANK	10.49%	16.18%	0.31	0.29	-22.50%	2020-03-05	2020-03-23	2020-10-27	0.47
LT	11.74%	17.28%	0.37	0.32	-37.82%	2020-01-30	2020-03-24	2020-12-16	0.31
TATAMOTORS	6.91%	22.72%	0.13	0.12	-51.66%	2020-02-05	2020-04-03	2020-11-18	0.13
BAJFINANCE	14.50%	17.90%	0.50	0.44	-36.32%	2020-03-03	2020-05-26	2020-11-18	0.40
ADANIENT	12.76%	35.22%	0.34	0.31	-63.63%	2023-01-09	2023-02-27	2024-05-31	0.20

Each sleeve begins with INR 20 million / 12; 'not yet' means unrecovered by the end date. Exact figures are in outputs_real/per_name_summary.csv.

Reconciled P&L decomposition

P&L component	Portfolio INR
Option premiums	INR 18,293,374
Stock delivery + mark	INR -3,055,018
Dividends	INR 906,295
Rights value	INR 73,736
Spun-off stock	INR 1,087,185
Cash interest	INR 0
All costs	INR -517,736
Open option liability	INR 0
Ending equity (incl. INR 20m start)	INR 36,787,835

Realized stock delivery P&L was INR 3,972,267; it is a subset of the stock line, which also marks remaining shares to the final close. Adding it again would double-count returns.

3. Data and methodology

The downloader checked weekdays from 2020-01-01 through 2026-06-30. It found 1,604 price dates, 1,603 option dates, and 4,891,323 option rows. 91 weekdays had all three NSE reports missing; one date had prices but no options report.

Source inventory

Dataset	Granularity and use	Known deficiency
NSE equity/index bhavcopy	Daily closes for 12 stocks and NIFTY 50	No intraday path or executable close quote
NSE F&O bhavcopy	Daily contract close, settlement, volume, OI and lot fields	No bid/ask or Greeks; many zero-volume rows
NSE corporate-action register	Ex-date dividends, splits, bonuses, rights and demergers	Audit cannot prove every small distribution is captured
NSE Indices NIFTY 50 TRI	Official daily total-return benchmark	Benchmark is investable only through a tracking vehicle

Universe selection

The original model defined a fixed 12-name F&O shortlist spanning energy, banks, IT, industrials, autos, and NBFCs. TCS represented the lower-volatility sleeve; TATAMOTORS and ADANIENT supplied cyclical and idiosyncratic stress. We retained this qualitative choice and tested option participation on 2020-01-01 before the first strategy entry. Eligibility required at least 250 option contracts traded, 1 million summed share open interest, and 15 rows with both volume and open interest; all 12 passed. Exact readings are in data/universe_selection.csv.

Assignment gap. This checks liquidity only within an already chosen shortlist. It does not satisfy the brief's requested objective selection from the full F&O candidate pool, and the fixed names have survivorship bias. A compliant new universe needs an ex-ante all-market screen and a rebuilt backtest; the present performance should be read with that limitation.

Coverage check	Value
Price dates	1,604
Option dates	1,603
Option rows	4,891,323
Duplicate contract keys	0
Price cells missing	0
Missing bid/ask fields	4,891,323 / 4,891,323
Zero volume fields	3,143,833

Execution uses traded option closes and prior-day stock closes for strike choice. Lot sizes come from NSE fields or checked legacy inference. Dividends and corporate events are applied on ex-dates; short-option sale STT and share-delivery STT are charged. Puts are fully cash-secured and calls share-covered. The 20%/35% SPAN-plus-delivery proxy is checked daily against cash and haircut share collateral. Full assumptions are machine-readable in outputs_real/assumptions.json.

Transaction-cost model

Cost item	Applied rule
Brokerage	INR 20 for every executed option, futures or delivery leg
Option exchange / tax	0.035% of premium exchange charge; 18% GST on brokerage + exchange; SEBI INR 10/crore
Option-sale STT	0.050% before Apr-2023; 0.0625% to Sep-2024; 0.100% to Mar-2026; 0.150% thereafter
Option slippage	25 bps of premium in the base case; 10 and 50 bps sensitivity cases
Physical stock delivery	0.1% STT plus delivery charges; mandatory strike delivery has no slippage
Market stock sales	Delivery costs plus 25 bps of notional for residual and spun-off-share sales
Futures legs	0.00173% exchange charge, GST, SEBI fee, 2 bps notional slippage; buy stamp duty 0.002%
Futures-sale STT	0.010% before Jun-2023; 0.0125% to Sep-2024; 0.020% thereafter
Futures roll	Charged as two executions: sell expiring contract and buy next contract

Cumulative modeled costs are INR 517,736 for Part A and INR 494,197 for Part B. Option exercise STT is borne by the purchaser and is therefore not charged to this short-option writer; delivery STT is charged where shares move.

For splits and bonuses, held share and open-option quantities scale by the exchange factor while strike, premium and cost basis scale inversely. The rights event uses the NSE strike and lot adjustment and credits modeled rights value; demergers create a separate spun-off claim, provisionally marked then sold at its first NSE listing close. Recorded cash dividends are credited to prior-close holdings on the ex-date. Expiry resets and symbol changes are recorded in the action log.

Corporate-action screen for all twelve names

Ticker	Recorded event(s)	Largest adjusted 1-day drop
RELIANCE	2020-05-13 RIGHTS; 2023-07-19 EXPIRY_RESET; 2023-07-20 SPIN_OFF; 2024-10-28 BONUS	13.15% (2020-03-23)
HDFCBANK	2025-08-26 BONUS	12.61% (2020-03-23)
ICICIBANK	No large action found	17.85% (2020-03-23)
TCS	No large action found	9.41% (2020-03-12)
INFY	No large action found	10.04% (2020-03-23)
SBIN	No large action found	14.4% (2024-06-04)
AXISBANK	No large action found	27.91% (2020-03-23)
KOTAKBANK	2026-01-14 SPLIT	13.0% (2020-03-23)
LT	No large action found	16.27% (2020-03-23)
TATAMOTORS	2025-10-13 EXPIRY_RESET; 2025-10-14 SPIN_OFF; 2025-10-24 SYMBOL_CHANGE	14.36% (2020-03-23)
BAJFINANCE	2025-06-16 SPLIT_BONUS	23.23% (2020-03-23)
ADANIENT	2025-11-17 RIGHTS	28.2% (2023-02-01)

The screen compares adjacent raw closes after applying recorded entitlements. It can identify large discontinuities; it cannot establish that every small corporate distribution is captured. Cash dividends are extracted separately from the NSE action register.

Interpretation limits. Bid and ask quotes were absent, so entry and mark prices use traded close and settlement fields. The 104 dated dividend records in data/dividends.csv are credited to shares held before each ex-date session. Rights are monetized at NSE theoretical values; spun-off shares use the special pre-open implied value until listing, then NSE closes. Margin is a proxy, not a historical exchange SPAN replay; see outputs_real/daily_positions.csv. Performance is exploratory, not a production claim.

4. Sensitivity and stress narrative

Scenario	Strike OTM	Slippage	CAGR	Vol	Sharpe	MDD
base	5.00%	25 bps	9.84%	12.05%	0.32	-27.74%
strike_3pct	3.00%	25 bps	9.36%	12.21%	0.28	-27.32%
strike_8pct	8.00%	25 bps	7.25%	10.98%	0.11	-27.01%
slippage_10bps	5.00%	10 bps	9.86%	12.05%	0.32	-27.73%
slippage_50bps	5.00%	50 bps	9.81%	12.05%	0.31	-27.74%

The strike choice matters: moving from 5% to 3% OTM changes Sharpe from 0.32 to 0.28; moving to 8% changes it to 0.11. The 10-50 bps slippage range changes CAGR by only about 0.05 percentage points, but that narrow modeled range cannot validate fills when every bid/ask field is missing.

The wheel's worst drawdown was -27.74%, from 2020-02-13 to 2020-03-23, with recovery on 2020-11-10. Cash-secured short puts were already open at the peak. At the 27 February expiry, the rules forced two assignments, in TATAMOTORS and LT; both then sold covered calls. Stock exposure rose from INR 982,350 to INR 1,928,676. The five largest sleeve losses and endpoint positions are below.

Sleeve	Peak position	Trough position	Stock move	Equity change
AXISBANK	0 shares; short put 690	0 shares; short put 700	-58.73%	INR -864,029
TATAMOTORS	0 shares; short put 175	8600 shares; short call 170	-60.94%	INR -845,731
LT	0 shares; short put 1300	1125 shares; short call 1300	-43.97%	INR -612,074
ICICIBANK	0 shares; short put 505	0 shares; short put 490	-47.50%	INR -521,568
BAJFINANCE	0 shares; short put 4100	0 shares; short put 4500	-52.74%	INR -507,060

AXISBANK lost INR -864,029 without holding shares at either endpoint: its open short put was marked against a -58.73% underlying move. TATAMOTORS and LT lost through delivered shares after the two assignments. The trade log shows 12 new option sales and INR 4,952 of logged event costs during this episode. Daily option liabilities, cash and shares are in outputs_real/daily_positions.csv.

5. Part B: implemented synthetic NIFTY wheel

NIFTY options settle in cash, so an ITM put cannot deliver an index basket. This implementation sells monthly cash-secured puts and, after an ITM expiry, buys the nearest monthly NIFTY futures contract on the next session. It then writes calls against the same number of index units. An ITM call settles in cash and closes the futures; an OTM call leaves the futures in place and triggers a roll at futures expiry. Calls cannot be struck below the recovery basis. Futures are marked to NSE settlement daily, transaction costs and slippage are charged on every futures leg, and lot changes preserve index units rounded down to a whole new lot.

Series	End value	CAGR	Vol	Sharpe	Sortino	MDD	Calmar
Synthetic NIFTY wheel	INR 27,527,240	5.04%	12.47%	-0.06	-0.05	-24.74%	0.20
NIFTY 50 total return	INR 42,283,722	12.22%	17.97%	0.38	0.35	-38.27%	0.32

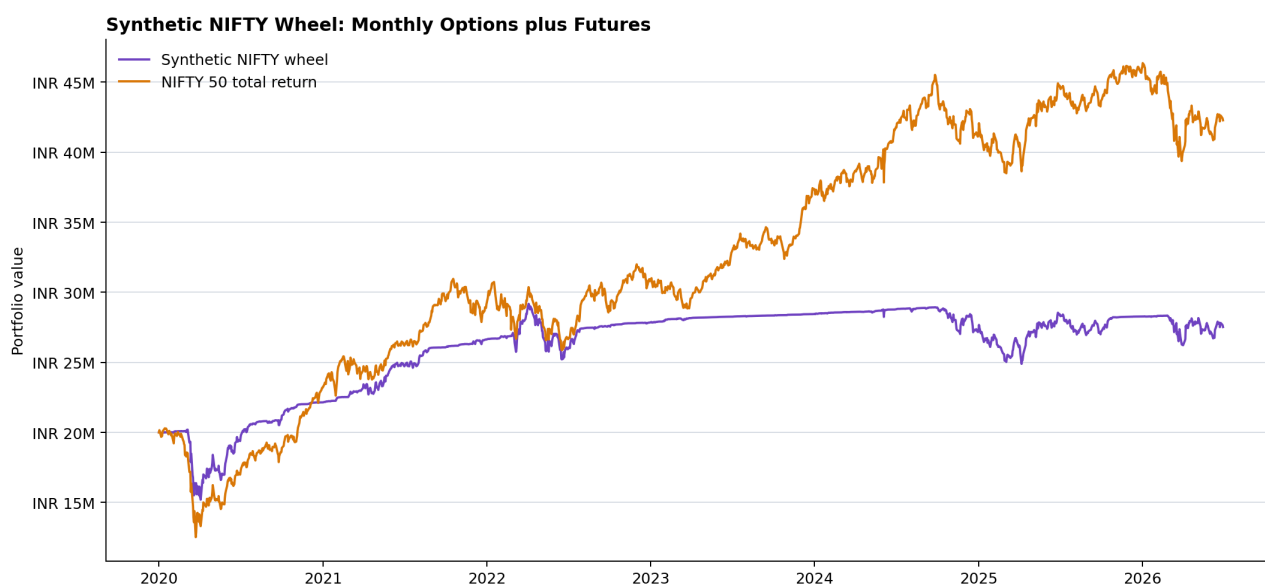


Figure 3. Implemented monthly synthetic NIFTY wheel versus NIFTY 50 total return.

Part B decision logic

Decision	Rule used by the engine
Contract set	NIFTY option expiries matching listed futures expiries; weekly options excluded
Reference / strike	Previous-day NIFTY close; put near 95%, call at or above max(105%, recovery basis)
Execution / mark	Option and futures entry at current close; options marked to settlement
Synthetic assignment	ITM put pays intrinsic; next session buys futures preserving index units
Recovery basis	Futures entry + put intrinsic per unit - original put premium per unit
Futures accounting	Daily settlement variation margin is credited/debited directly to cash
Call-away / roll	ITM call pays intrinsic and closes futures; OTM call retains and rolls futures
NAV / cash	Cash - short-option liability; futures P&L is already in cash; zero cash interest

Across 2020-01-01 to 2026-06-30, the engine sold 48 puts; 12.50% settled ITM and initiated futures exposure. It sold 30 calls, with 16.67% ending in a synthetic call-away, and completed 24 futures rolls. It recorded 15 stale option marks and 1 stale futures marks, which retain the last settlement. The account reserves the full put strike obligation; while futures and a short call are held it uses the larger of 15% of futures notional and 20% of call notional as a documented margin proxy. Cash posted as security or margin earns no interest in the Part B base case.

Part A versus Part B mechanics and risk

Issue	Stock wheel	Synthetic NIFTY wheel
Assignment	Physical shares move at strike	Option pays cash intrinsic; futures bought next session
Primary risk	Market beta plus single-name gaps and corporate events	Diversified index beta, but concentrated market-crash exposure
Basis / timing	Option and delivered stock share one underlying	Expiry spot settlement versus next-session futures entry creates gap and basis risk
Roll	No stock roll after delivery	24 futures rolls; each pays two legs; cumulative raw calendar spread 1,804.15 index points
Margin offsets	Cash-secured put; covered stock supports call after 20% haircut	Uses max(15% futures, 20% call), assuming an offset; actual historical SPAN may differ
Lot granularity	Each stock sleeve rounds independently to its exchange lot	One index lot controls all units; lot revisions round exposure down to a whole new lot
Expiry choice	Nearest monthly with at least 7 DTE	Monthly expiries aligned to futures; weekly would add turnover and near-expiry gamma

Futures basis normally converges toward spot at expiry, but this engine can still lose from the gap between option settlement and the next-session futures purchase, and from the price difference between expiring and next-month futures at a roll. Margin offsets reduce the proxy requirement rather than eliminating economic leverage. The index removes company-specific delivery shocks, while leaving the portfolio exposed to a common market sell-off across every constituent.

Part B ledger and sensitivity

Reconciled component	INR
Starting capital	INR 20,000,000
Option premium	INR 6,866,404
Option cash settlement	INR -8,840,078
Futures variation margin	INR 9,995,112
Cash interest	INR 0
Costs	INR -494,197
Ending equity	INR 27,527,240

Scenario	OTM	Opt slip	Fut slip	CAGR	Sharpe	MDD
base	5.00%	25 bps	2 bps	5.04%	-0.06	-24.74%
strike_3pct	3.00%	25 bps	2 bps	6.59%	0.07	-29.04%
strike_8pct	8.00%	25 bps	2 bps	4.77%	-0.10	-27.23%
option_slippage_10bps	5.00%	10 bps	2 bps	5.05%	-0.06	-24.74%
option_slippage_50bps	5.00%	50 bps	2 bps	5.03%	-0.06	-24.74%
futures_slippage_4bps	5.00%	25 bps	4 bps	4.89%	-0.07	-24.75%

The base case ended below NIFTY total return but with lower volatility and a materially smaller maximum drawdown. The 3% OTM case improved return while deepening drawdown; 8% OTM reduced both premium income and return. Futures slippage matters more than option-premium slippage because it is charged on full futures notional.

The option expiry cash debit uses intrinsic value calculated from the expiry-day NIFTY close because the historical archive does not supply the official final-settlement index as a clean separate field throughout the sample. Entry uses the next session's observed close, not its open, and historical SPAN files and executable bid/ask quotes are unavailable. These are material implementation limits. Weekly expiries were excluded by matching option expiries to listed NIFTY futures expiries; a weekly version would have higher turnover and near-expiry gamma exposure.

6. Assignment coverage, limitations and reproducibility

Assignment item	Report evidence	Status
2020-01-01 to 2026-06-30 EOD data	Section 3 source inventory and coverage audit	Covered
Objective 8-15 stock liquidity screen	12 names pass inception liquidity thresholds, but shortlist was chosen first	Gap
Put, delivery, call, sizing, exits and margin	Section 2 rules, decision table and worked RELIANCE example	Covered
Costs, STT and slippage sensitivity	Section 3 cost table; Section 4 sensitivity	Covered
Portfolio and per-name metrics / diagnostics	Sections 1-2 tables and reconciled P&L	Covered
Benchmarks and worst-drawdown narrative	NIFTY 50 TR, universe buy-and-hold, and March 2020 walk-through	Covered
Synthetic-index design and comparison	Section 5 mechanics, basis, roll, margin, lots and risk comparison	Covered
Biases, sources, verdict and reproducibility	Section 6 and machine-readable output files	Covered

Daily equity and F&O bhavcopies come from NSE reports; cache paths and missing dates appear in data/source_manifest.json, while archive URL patterns are in main/data_pipeline/fetch_nse_data.py. Event links are in data/corporate_actions.csv. Benchmark: NSE Indices historical TRI. Tax: NSE STT rates and payer. Margin: NSE margin FAQ. The 20%/35% coefficients are proxies, not daily exchange SPAN calculations. Unlisted strikes and stale option marks are possible; bid/ask fields are absent. Previous-day stock prices avoid strike-selection look-ahead, but filling at the same day's observed option close remains optimistic. Fixed 2026-observable names introduce survivorship bias.

Deployability verdict. The stock wheel returned 9.84% annualized and the synthetic NIFTY wheel returned 5.04%, versus 12.22% for NIFTY 50 total return. Neither implementation credits interest on option security or futures-margin cash. The evidence is not sufficient for a production deployment claim until execution prices and historical margins are tested with more realistic data and the Part A universe is selected objectively.

Run `python3 -m main.data_pipeline.fetch_nifty_tr`, `python3 -m main.data_pipeline.extract_nifty_derivatives`, `python3 -m main.backtest.run_real_backtest`, `python3 -m main.backtest.plot_real_results`, `python3 -m main.backtest.run_nifty_wheel`, and finally `python3 -m main.report.build_report`. Tests are isolated under tests/. Install requirements.txt first.